

5-Bar SCARA v2 Onboarding Package

M6

Encoder + Sensor Fusion Edition | Engineering Roadmap

SYSTEMS & VALIDATION LEAD

1 | ROLE & OBJECTIVE

As **Systems & Validation Lead (M6)**, you have the widest scope on the team. You own the **Pre-Phase baseline audit** — establishing the verified Git foundation before any code changes — and you co-own **Phase 6** with M5, delivering the watchdog timer configuration, the final performance specification documentation, and overseeing the 30-cycle acceptance test. Between those bookends, you serve as **support in Phase 5** and as the team's **escalation authority** and cross-team integrator.

Primary Mission: Create a clean, documented baseline that every other member can build upon; configure system-level fault protection (WDT, stall timer); author the final performance specification; and ensure the project closes with a signed, witnessed 30-cycle acceptance test.

2 | HARDWARE & COMPONENTS REQUIRED

Component	Purpose	Notes
Both ESP32 + ESP32-CAM boards	Pre-Phase: verify existing firmware runs before any modification	Flash and test BEFORE any member modifies code
Full wiring harness (as-built)	Pre-Phase: photograph and label every wire; create GPIO pin table	Critical for rollback if v2 changes cause faults
Git repository (remote + local)	Version control custodian; tag v0.0-baseline before Day 2	GitHub / GitLab; enforce branch protection
USB-to-Serial adapter	Flash and Serial-monitor both nodes during baseline verification	2 adapters recommended
Physical alignment tape/marker	Mark world-frame origin and home position before M1 modifies frame	Photographed and documented in baseline
Oscilloscope (borrowed from M5)	WDT and stall timer verification in Phase 6	Verify 50ms timer period
Documentation tools (laptop)	Write performance spec table; author sign-off sheet; maintain test log master	Shared Google Doc or Confluence recommended

3 | SOFTWARE & PLATFORMS REQUIRED

Tool	Purpose	Source
Git (CLI + GitHub Desktop)	Create v0.0-baseline tag; manage branches; resolve merge conflicts	git-scm.com
Arduino IDE 2.x / PlatformIO	Flash and verify baseline firmware on both nodes	arduino.cc
config.h editor (VSCode)	Create config.h skeleton with encoder PPR field placeholder	code.visualstudio.com
Markdown editor (Typora / VSCode)	Write pre-phase audit report, performance spec, sign-off sheet	typora.io
Python + pandas	Aggregate test logs from M5; generate final performance table	pip install pandas
Spreadsheet	GPIO pin table; Phase 0 audit checklist; performance spec matrix	Excel / Google Sheets
Camera (smartphone)	Photograph wiring harness, GPIO table, world-frame origin marker	Any smartphone

4 | AI INTEGRATION STRATEGY

Your work is systems-level: documentation quality, configuration management, and fault-protection code. Use AI to generate audit templates, WDT configuration, and final specification documents:

Prompt 1 — Generate GPIO audit table:

"I have an ESP32 connected to: DRV8825 stepper driver (STEP, DIR, ENABLE pins), encoder 1 (A, B channels on interrupt-capable pins), encoder 2 (A, B channels), ESP-NOW (WiFi, no extra GPIO), relay module (digital output), and UART Serial. Create a complete GPIO pin assignment table showing: GPIO number, function, direction (IN/OUT), interrupt-capable (Y/N), and notes on constraints. Flag any conflicts or pins that cannot be used as interrupts on ESP32."

Prompt 2 — Watchdog timer configuration:

"I need to configure esp_task_wdt on an ESP32 Arduino project for a robot pick-and-place system. The WDT timeout should be 3 seconds. I also need a hardware timer ISR at 50ms for stall detection. Provide: (1) complete WDT init code with esp_task_wdt_init, esp_task_wdt_add, and esp_task_wdt_reset placement, (2) hardware timer setup with timerBegin/timerAttachInterrupt/timerAlarmWrite for 50ms, (3) explain what happens if WDT fires and how to recover."

Prompt 3 — Performance specification template:

"Generate a professional performance specification table for a robotics project with these parameters: Vision pipeline latency (target <100ms), ESP-NOW latency (target <15ms), IK solve time (target <1ms), Encoder correction latency (target <200ms), Stall detection response (target <700ms), Kalman position noise stddev (target <1mm), Pick success rate (target >=93%), Packet loss rate (target <2%), Full cycle time (target <9s). Format as a formal table with columns: Parameter | Target | Measured | Pass/Fail. Include a project sign-off block at the bottom with signature lines for 6 team members."

Prompt 4 — Git branching strategy for the team:

"We have 6 team members working on a microcontroller firmware project with 2 firmware projects (kinematic node and vision node). We need: feature branches per phase, hard gate merges only after PASS sign-off, version tags at each phase completion, and protection of the main branch. Design a Git branching strategy and naming convention. Include the exact git commands to create the v0.0-baseline tag, create a feature branch, and merge after gate PASS."

5 | BASIC ARCHITECTURE & CONNECTIONS

Your Deliverable	Who Depends On It	Dependency Type
v0.0-baseline Git tag + config.h skeleton	ALL 6 MEMBERS	Every member starts from this baseline — BLOCKING
GPIO pin table (4 encoder pins reserved)	M1 (mounting) + M2 (ISR)	Wrong pin = ISR fails silently; CRITICAL
World-frame origin photograph + marker	M1 (measurements) + M4 (calibration)	IK coordinate reference; must precede Phase 1
Pre-Phase PASS confirmation to M2	M2 cannot begin Phase 2 until M6 signs Pre-Phase PASS	Hard gate; no exceptions
WDT configuration (Phase 6)	Both ESP32 nodes (M2 + M3 firmware)	System-level fault protection for unattended run
Performance Spec v2 final table	M5 + all members (sign-off sheet)	Project closure documentation; required for sign-off
Escalation arbitration	Any member blocked by inter-team dispute	Authority to override if deadlocked

Your Scope: Pre-Phase → [all phases run] → Phase 5 Support (integration observer) → Phase 6: WDT + hardware hardening → 30-cycle acceptance → Performance Spec → Sign-off

6 | TIMELINE & MILESTONES

[DAY 1 — PRE-PHASE (CRITICAL PATH)]

- Record ALL existing AccelStepper maxSpeed() and setAcceleration() values from current firmware
- Photograph full wiring harness — label every wire with GPIO number and function
- Create GPIO pin table: identify which 4 interrupt-capable pins are reserved for encoders
- Verify recommended encoder GPIO: 34 (ENC1_A), 35 (ENC1_B), 36 (ENC2_A), 39 (ENC2_B)
- Create config.h skeleton with ENCODER_PPR placeholder; commit to Git
- Tag current codebase: git tag v0.0-baseline && git push origin v0.0-baseline
- Confirm M2 verifies robot moves under existing firmware — obtain M2 sign-off
- Pre-Phase PASS criteria: baseline tagged, 4 GPIO reserved, M2 confirms movement

[DAYS 2–13 (Monitor & Support Role)]

- Attend daily standups; track each phase's gate status on master checklist
- Resolve any Git merge conflicts or config.h update disputes immediately
- Phase 5 Support: be present during integration; assist M4 with calibration logistics
- Maintain master test log: collect M5's signed test results after each phase gate
- Flag to all members if any HARD GATE is at risk of blocking downstream phases

[DAYS 14–16 — PHASE 6 CO-OWNERSHIP]

- Configure WDT on ESP32 (kinematic): esp_task_wdt_init(3, true) + reset in main loop
- Configure WDT on ESP32-CAM (vision): same 3-second WDT around camera loop
- Verify 50ms hardware stall timer: timerBegin + timerAttachInterrupt + timerAlarmWrite
- Oversee M5's power hardening: 1000µF DRV8825, 100nF VCC and encoder, ferrite bead
- Author Performance Specification v2 table: all parameters with measured values from M5's logs

- Coordinate 30-cycle acceptance test scheduling — all 6 members must be present
- Witness 30-cycle run; document each cycle outcome (success/fail/stall/occlusion)
- Confirm PASS 1–6 criteria met; circulate sign-off sheet; collect all 6 signatures

7 | ESCALATION MATRIX

You ARE the top-level escalation point. Below are the situations where you should proactively intervene, and your recommended actions:

Situation	Your Action	Timeframe
Pre-Phase: any member starts modifying code before baseline is tagged	Stop modification immediately; complete baseline first	Day 1 — BLOCKING
Phase gate claimed PASSED without M5 written sign-off	Reject advancement; request documented test log	Immediate
Phase 2B and 3B both delayed: Phase 4 at risk	Re-prioritize team capacity; identify if parallel work is possible	Flag by Day 8
Inter-member dispute on config.h values or packet format	Arbitrate: use original spec + AI validation; make final call	Within 24 hours
30-cycle run fails PASS 2 (unrecoverable fault)	Document failure; do NOT re-run without root cause fix	Immediate stop
WDT fires during integration testing	Log stack trace; coordinate M2/M3 fix; verify reset recovery	Before Phase 6 acceptance
Any member unavailable for phase gate review	Assign temporary reviewer from remaining team with M6 oversight	Day of gap

You set the foundation everyone builds on and close the loop everyone depends on. The team's success starts with your Day 1 baseline commit and ends with your Phase 6 sign-off. Lead with clarity, enforce the gates, and drive the team home. ■